
The Toolkit

UTSA Problem Solving Club • Fall 2026

The Fly and the Trains

Two trains on the same track, 200 miles apart, head toward each other at 50 mph each. A fly starts at the front of one train, flies to the other at 75 mph, instantly turns around, flies back, and keeps zig-zagging until the trains collide.

How far does the fly travel in total? Find the answer in two or more different ways.

Apocryphal: John von Neumann answered instantly by a harder method unaware of an easier shortcut.

A Geometric Series in Disguise

Prove that for every integer $n \geq 0$ and every θ with $0 < \theta < 2\pi$,

$$1 + \cos \theta + \cos 2\theta + \cdots + \cos n\theta = \frac{1}{2} + \frac{\sin((n + \frac{1}{2})\theta)}{2 \sin(\theta/2)}.$$

Hint: Then take the real part of the sum the geometric progression $1 + z + z^2 + \cdots + z^{n-1} = (z^n - 1)/(z - 1)$ with $z = e^{i\theta} = \cos \theta + i \sin \theta$. *Bonus:* What identity do the imaginary parts give you?

Roots You Never Have to Find

Let a, b, c be the three roots of $x^3 - 5x + 3 = 0$. Compute **(a)** $a + b + c$, $ab + bc + ca$, abc ; **(b)** $a^2 + b^2 + c^2$; **(c)** $1/a + 1/b + 1/c$; **(d)** $a^3 + b^3 + c^3$; **(e)** $a^4 + b^4 + c^4$.

Hint for (d) and (e): Each root satisfies $a^3 = 5a - 3$.

Keep your primes close, and your coprimes closer

(a) How many positive integers $n \leq 1,000,000$ are *not* divisible by 2? By neither 2 nor 3? By neither 2 nor 3 nor 5? **(b)** A day has 86,400 seconds. How many positive integers $n \leq 86,400$ are *not* divisible by 2? By neither 2 nor 3? By neither 2 nor 3 nor 5? **(c)** How does the fraction of numbers left over in part (b) from the total 86,400 compare to the fraction of numbers left over in part (a) from the total of 1,000,000? Are those fractions exactly the same? Why? **(d)** If n is a large integer, how can you predict (almost exactly, up to a very small error) the fraction of positive integers $\leq n$ that are *not* divisible by, say, three given primes p, q, r ? For what values of n is your prediction exact? How far off can your prediction deviate from the exact answer otherwise?

Too many zeros!!! Plus: one-fourth of a mystery

(a) In how many zeros does one-million factorial $1,000,000! = 1 \cdot 2 \cdot 3 \cdots 999,999 \cdot 1,000,000$ end? **(b)** The answer to (a) turns out to be a little less than $250,000 = \frac{1}{4} \cdot 1,000,000$. Can you explain why?

Competition Caliber

Spreading a Fixed Total

Let A be a positive real number. What are the possible values of $\sum_{j=0}^{\infty} x_j^2$, given that $x_0, x_1, \dots$ are positive numbers for which $\sum_{j=0}^{\infty} x_j = A$?

Source: Putnam 2000 A1

Every Side and Every Diagonal

A regular n -gon is inscribed in the unit circle. Consider all segments joining pairs of vertices—all sides and all diagonals. **Find the product of all these segment lengths.**

Help: First find the product of the distances from one fixed vertex to all of the others.

Six Functions, One Minimum

Find the minimum value of

$$|\sin x + \cos x + \tan x + \cot x + \sec x + \csc x|$$

for real numbers x (wherever the expression is defined).

Worth trying: Let $s = \sin x + \cos x$. Every other term is a rational function of s .

Source: Putnam 2003 A3

A Polynomial with Few Terms

Let k be the smallest positive integer for which there exist distinct integers m_1, m_2, m_3, m_4, m_5 such that the polynomial

$$p(x) = (x - m_1)(x - m_2)(x - m_3)(x - m_4)(x - m_5)$$

has exactly k nonzero coefficients. Find, with proof, a set of integers m_1, m_2, m_3, m_4, m_5 for which this minimum k is achieved.

Source: Putnam 1985 B1

Unit Fractions

Find all ordered pairs (a, b) of positive integers for which

$$\frac{1}{a} + \frac{1}{b} = \frac{3}{2018}.$$

Source: Putnam 2018 A1

Three Sets, Ten Elements

Determine, with proof, the number of ordered triples (A_1, A_2, A_3) of sets which have the property that (i) $A_1 \cup A_2 \cup A_3 = \{1, 2, \dots, 10\}$, and (ii) $A_1 \cap A_2 \cap A_3 = \emptyset$. Express your answer in the form $2^a 3^b 5^c 7^d$.

Source: Putnam 1985 A1